Joan Guerrero Sanchez

Software Developer | AI Specialist (DAM + IA) Barcelona, España
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PROFESSIONAL PROFILE Full Stack Developer specializing in Artificial Intelligence and Big Data. I combine a solid foundation in software development (Java, Python) with practical experience in Deep Learning and RAG architectures. Currently completing my DAW (Digital Architecture) degree and seeking to apply innovative technical solutions. Proven ability for self-learning and developing complex AI projects from scratch.

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), Java, JavaScript (ES6+), TypeScript, SQL, Bash.
- **Artificial Intelligence & Data:** PyTorch, TensorFlow, Hugging Face Transformers, RAG (Retrieval-Augmented Generation), LangChain, FAISS, Computer Vision (OpenCV), Pandas, NumPy.
- **Web & Mobile Development**: React, Vue.js, Node.js (Express), React Native, HTML5, CSS3/Sass.
- **Tools & DevOps:** Git, GitHub, Docker, Linux, Agile Methodologies (Scrum).
- **Databases:** MongoDB, SQL (MySQL, PostgreSQL).

ENGINEERING & AI PROJECTS

Local RAG Document Assistant | *Python, Llama-3, FAISS, Gradio*

- Development of a local Artificial Intelligence system capable of processing and "chatting" with private PDF documents without sending data to the cloud.
- Implementation of data ingestion, chunking, and semantic vector search pipelines using FAISS and Sentence Transformers.
- Optimization of inference through quantization of LLM models (Llama-3, Phi-3) for efficient execution on consumer hardware.

Doom Autonomous Agent (Reinforcement Learning) | *Python, PyTorch, Stable Baselines 3*

- Training of an autonomous agent capable of navigating and fighting in the 3D environment VizDoom using PPO (Proximal Policy Optimization) algorithms.
- Design of a modular software architecture applying the Strategy Pattern to manage dynamic reward functions.
- Implementation of Curriculum Learning to improve model convergence in progressively challenging scenarios.

BERT Sentiment Classifier | *PyTorch, Transformers*

- Fine-tuning of a BERT (Bidirectional Encoder Representations from Transformers) model for sentiment classification on massive datasets.
- Creation of a custom neural network on the base architecture to optimize accuracy in specific domains.

Web Platform for a Gaming Community | *Full Stack (React, Node.js, SQL)*

- Design and deployment of a complete web platform with a secure authentication system (JWT/OAuth).
- Optimization of a relational database for real-time user ranking management.

WORK EXPERIENCE

Back End Developer | Myllox | *February 2026 - Present*

- I contributed to the Node.js server runtime (Express and ES modules): a multi-process architecture with a process monitor and specialized services, IPC communication, and an administration API over HTTPS.
- Integration with a centralized server (JWT authentication/sessions), configuration support for Nginx (reverse proxy), and service health monitoring with alerts.
- Development of modular JavaScript operational interfaces (server/service/runtime dashboards): real-time indicators, ping history, log panels, and polling strategies for consecutive failures.
- Automated API testing with Jest and Supertest.

Front-End Developer (Internship) | *Myllox | March 2024 - June 2024*

- I led the adaptation of the web platform to a mobile environment using React Native.
- I improved accessibility and the user interface (UI/UX), ensuring smooth performance across various devices.

L1 Logistics Associate | *Amazon | June 2025 - February 2026*

- Management of logistics processes, meeting strict performance and quality objectives.
- Development of problem-solving skills under pressure and teamwork in an advanced technological environment.

ADDITIONAL EXPERIENCE

- Previous experience in the service and customer service sector (Burger King, Goiko), developing time management and customer service skills.

ACADEMIC BACKGROUND

- **Advanced Technician in Web Application Development (DAW)** | *Institut TIC de Barcelona | 2025 - 2026*
- **Specialization Course in Artificial Intelligence and Big Data** | *Institut TIC de Barcelona | 2024 - 2025*
- **Advanced Technician in Multiplatform Application Development (DAM)** | *Instituto Pedralbes | 2022 - 2024*

LANGUAGES

- **Spanish**: Native
- **English**: Professional (B2/C1).
- **German:** Basic (A2)
